

An Exploratory Study of Chewing-Synchronized Intraoral Electrotactile Stimulation for Modulating Texture-Related Perception During Mastication

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Abstract—Food texture perception emerges from the integration of oral tactile feedback, jaw movement, and other sensory information during mastication. Although prior work has explored food texture augmentation through auditory, mechanical, and muscular feedback, the role of intraoral tactile feedback during active chewing remains underexplored. This paper investigates whether chewing-synchronized gingival electrotactile stimulation can influence texture-related perception during mastication. We developed a low-latency system that detects chewing phases in real time and delivers electrical stimulation to the maxillary and mandibular gingiva near the molars in response to detected occlusion or mouth opening. In a within-subjects study, six participants chewed a food-grade silicone tube and a gummy candy under no-stimulation and electrotactile stimulation conditions at 200 Hz or 1000 Hz. Participants rated texture-related perceptions. The results suggest that the interpretation of the evoked oral sensation varied across stimulation conditions and chewing phases. In the silicone tube condition, stimulation triggered by mouth opening was associated with higher viscosity/adhesiveness ratings, whereas stimulation triggered by occlusion was associated with higher crunchiness ratings. In the gummy candy condition, fewer texture-related differences were observed, although elasticity showed an exploratory condition-dependent pattern. These findings provide initial evidence that chewing-synchronized gingival electrotactile feedback may influence the interpretation of texture-related impressions when embedded in the sensorimotor context of mastication.

Index Terms—Food texture perception, intraoral haptics, electrotactile stimulation.

I. INTRODUCTION

FOOD texture plays an important role not only in hedonic eating experiences but also in the control of mastication. According to ISO, food texture is perceived through mechanical, tactile, and, where appropriate, visual and auditory cues [1]. Food texture can therefore be understood as a multisensory percept that emerges from the integration of sensory information with contextual factors and prior experience. During mastication, this percept is continuously updated through temporally coupled signals arising from intraoral tactile feedback, muscle feedback, visual information, and sound [2], [3]. Because these signals occur simultaneously and interact dynamically, isolating the contribution of a specific source of oral sensory feedback from others remains difficult, particularly for tactile feedback arising from intraoral tissues.

Among these sensory feedback channels, haptic information from the oral cavity and masticatory system is especially important because it directly contributes to detecting food

properties during chewing and to the control of mastication [2], [4], [5]. Somatosensory feedback during mastication includes kinesthetic feedback from the masticatory muscles and tactile feedback from periodontal and gingival tissues [4]–[9]. These somatosensory signals contribute to the perception of tooth loading and tooth contact, as well as to the regulation of bite force. However, the specific role of such intraoral tactile feedback in shaping texture-related perception during active chewing remains insufficiently understood.

Haptics and HCI research has explored ways to augment or render food texture by modulating sensory feedback during eating. Prior systems have manipulated different components of the eating experience, including auditory cues, mechanical resistance during mastication, electrical stimulation of the masticatory muscles, and other forms of sensory substitution [10]–[13]. These studies demonstrate that food-texture perception can be influenced by manipulating sensory and motor feedback during eating. However, most of these approaches do not directly manipulate tactile feedback arising from intraoral tissues during active chewing.

Beyond approaches aimed directly at altering food-texture perception, prior studies have investigated oral sensations by applying electrical stimulation of oral tissues. Electrical stimulation has been widely used because compact electrodes enable localized stimulation within the confined oral cavity, particularly for taste and electrotactile modulation of the tongue [14]–[16]. Electrical stimulation has also been applied to gingival and periodontal regions that are more directly involved in mastication [17]–[19]. However, these studies have primarily focused on the characteristics of the evoked haptic sensations themselves or their effects on mastication-related functions, with limited discussion of how such sensations contribute to food texture perception. Moreover, an interface used in prior work covered the occlusal surface, which may have altered the haptic interaction during chewing [17]. Other studies have examined intraoral sensations only in static conditions, without considering the dynamic context of mastication. These limitations suggest the need for an approach that can modulate intraoral tactile feedback while preserving natural tooth–food contact [18], [19].

To address this gap, this study investigates whether chewing-synchronized gingival electrotactile stimulation can modulate texture-related perception during active mastication. We developed a low-latency system that delivers controlled electrotactile stimulation to the maxillary and mandibular gingiva near the molar region while leaving the occlusal surfaces

uncovered, thereby preserving natural tooth–food contact. By synchronizing stimulation with different phases of chewing and varying the stimulation frequency, we examine how the temporal context of intraoral tactile feedback influences texture-related perceptual qualities during mastication.

We conducted a within-subjects study using a food-grade silicone tube and an edible gummy candy as chewing samples. Electrotactile stimulation at 200 or 1000 Hz was delivered during either occlusion or mouth opening, and participants evaluated texture-related perceptual qualities after each chewing condition. This design allowed us to examine how controlled intraoral tactile feedback is interpreted during ongoing mastication while preserving direct interaction between the teeth and chewing samples.

The contributions of this paper are as follows:

- We present a low-latency intraoral electrotactile stimulation system that detects chewing phases in real time and delivers phase-synchronized gingival stimulation while leaving the occlusal surfaces uncovered, allowing direct tooth–sample contact during chewing.
- Using this system, we provide exploratory evidence that phase-synchronized stimulation during chewing can influence specific texture-related subjective ratings.

II. RELATED WORK

This work is situated at the intersection of food texture perception, haptic augmentation of eating experiences, and intraoral electrotactile feedback. We first review food texture as a dynamic perceptual process formed through active mastication. We then review HCI and VR systems that modulate texture perception. Finally, we discuss intraoral electrical stimulation for generating oral tactile sensations, highlighting the gap addressed by this work.

A. Food Texture as a Dynamic Perceptual Process

Food texture has long been studied in relation to the mechanical and structural properties of food, such as hardness, adhesiveness, viscosity, elasticity, and particle structure [2], [20], [21]. These properties have commonly been characterized through instrumental measurements and related to sensory evaluations of texture. However, texture perception during eating is not determined solely by static material properties. Once food enters the mouth, its physical state continuously changes through fracture, deformation, saliva mixing, bolus formation, and repeated tooth–food contact [22]. Consequently, texture perception evolves throughout oral processing [3].

During mastication, jaw movements, tooth loading, and deformation of oral tissues continuously produce tactile, kinesthetic, and proprioceptive feedback. These sensory signals are interpreted in relation to ongoing motor activity and the evolving mechanical state of the food, forming texture percepts over time [3]–[5].

Importantly, these sensory events occur in a structured temporal relationship with the mastication cycle. For example, tooth loading and food compression occur around occlusion, whereas release and separation occur as the jaw opens. The perceptual contribution of tactile information may therefore

depend not only on the characteristics of the sensory signal itself, but also on when that signal occurs within the ongoing sensorimotor sequence.

Prior haptic studies have also suggested that tactile feedback can acquire different perceptual meanings depending on its temporal relationship to bodily actions or contextual events [23]. This suggests that, during mastication, the contribution of tactile information may depend not only on the characteristics of the sensory signal itself, but also on when it occurs within the chewing cycle. In this context, feedback delivered during biting or tooth–food contact may be associated with resistance, fracture, or structural change, whereas feedback delivered during mouth opening may be associated with separation, adhesion, or viscosity. Thus, the timing of stimulation is not only a technical parameter for stimulus delivery, but also a perceptual design factor that may shape how artificial oral tactile feedback is interpreted during eating.

Although prior studies in food science and oral physiology have shown that texture perception emerges through oral processing and sensory feedback, it remains difficult to determine how specific intraoral tactile cues contribute to texture perception during natural chewing. In ordinary eating, intraoral tactile feedback is inseparably coupled with jaw motion, muscle activity, auditory cues, and the changing mechanical state of the food. This coupling makes it difficult to manipulate oral tactile feedback independently while preserving the active sensorimotor context in which texture is normally perceived. Therefore, an intervention-based approach is needed to selectively modulate intraoral tactile feedback during mastication.

B. Sensory Interventions for Food Texture Augmentation

HCI and virtual reality research has explored ways to alter food texture perception without changing the physical food itself. A common strategy is to manipulate sensory cues that are naturally coupled with eating. For example, auditory feedback has been used to change perceived crispness and freshness by modifying chewing sounds [24]. Chewing Jockey extended this idea by detecting chewing actions and presenting augmented chewing sounds in real time, showing that texture perception can be altered through temporally synchronized auditory feedback [10].

Other systems have focused on mechanical or kinesthetic feedback. The Food Simulator presented force feedback to the teeth during biting, together with auditory and olfactory cues, to reproduce virtual food textures [11]. Jamming-based texture displays have controlled the stiffness of deformable objects to simulate different levels of hardness [25]. More recently, pneumatic interfaces such as VirCHEW Reality have provided chewing-related resistance or facial kinesthetic feedback to enhance virtual eating experiences [13]. These approaches show that mechanical resistance and chewing-related force cues are powerful components of food texture rendering.

Electrical muscle stimulation has also been used to intervene in the motor side of mastication. Nijima proposed a virtual food texture system that stimulates the masticatory

muscles and changes the perceived effort or resistance during chewing [12]. This approach directly modifies chewing-related kinesthetic sensation, but its primary target is muscle activation and jaw-related feedback rather than tactile feedback from oral tissues.

Overall, prior interfaces have shown that food texture perception can be modulated through auditory, mechanical, pneumatic, kinesthetic, and muscular pathways. However, these systems have manipulated cues outside the oral cavity, altered jaw-related feedback, or were unable to independently modulate intraoral tactile feedback. Thus, it remains unclear how localized artificial tactile feedback near the molar region contributes to perceived food texture during natural mastication while preserving tooth-sample contact.

C. Intraoral Electrotactile Feedback

Electrical stimulation is well suited for intraoral interfaces because it can be implemented in a compact form, precisely controlled, and used to evoke a variety of sensory percepts. Accordingly, electrical stimulation has been applied in the oral cavity for taste modulation, tactile display, sensory substitution, and clinical evaluation.

Prior systems have delivered electrical stimuli to the tongue, palate, lips, or eating utensils to evoke taste-like sensations, present electrotactile information, or support oral sensory and motor functions [14]–[16], [26]–[28]. These studies demonstrate the feasibility of electrical stimulation as an intraoral sensory interface.

Electrical stimulation has also been investigated as a means of generating and characterizing tactile sensations within the oral cavity. Electrical stimulation around the periodontal ligament has been shown to evoke tooth-related tactile sensation with perceptual responses varying across stimulation locations [19]. Gingival electrotactile stimulation near the molar region has further been characterized across stimulation frequencies and locations, showing that electrical stimulation can evoke localized oral sensations with different perceptual qualities [18].

Beyond the generation of intraoral sensations, electrical stimulation has also been incorporated into active chewing. Chewing-synchronized stimulation applied to the gum was shown to alter mastication force and other chewing parameters [17]. These findings suggest that intraoral electrical stimulation can both evoke tactile sensations and influence sensorimotor behavior during mastication. However, prior studies characterizing electrically evoked oral sensations were conducted under static conditions without mastication [18], [19]. Although these studies showed that intraoral electrical stimulation can evoke localized tactile sensations, they did not examine how such feedback affects texture perception during actual chewing. Closed-loop intraoral stimulation has been investigated during mastication, but this work focused on mastication force and used a mouthpiece that covered the occlusal surface [17]. Thus, it remains unclear whether intraoral electrotactile feedback can modulate texture-related subjective perception during actual chewing while preserving natural tooth-food contact.

III. SYSTEM DESIGN AND IMPLEMENTATION

We developed a real-time intraoral electrotactile stimulation system to investigate how temporally synchronized oral tactile feedback is interpreted during active mastication. The system detects mastication phases during chewing and delivers electrical stimulation to the gingiva in synchrony with the detected phase. The design goal was to provide temporally precise intraoral tactile feedback while ensuring electrical safety and minimizing physical interference with natural chewing.

A. System Architecture and Low-Latency Design

The central design requirement of our system was temporal alignment between mastication events and electrotactile feedback. Because the perceptual meaning of oral tactile feedback may depend on when it occurs during chewing, the system needed to detect mastication phases and deliver stimulation with minimal delay. We therefore designed the system as a wired, low-latency closed-loop trigger system that delivers electrical stimulation in synchrony with detected mastication events.

The system consisted of three components: a mastication detection module, a stimulation control module, and an intraoral electrode fixation module. The mastication detection module used either a pressure sensor or an inertial measurement unit (IMU), depending on the experimental condition. Sensor signals were processed by a microcontroller (Seeed Studio XIAO RP2040), which generated trigger signals for a programmable current stimulator (StimJim, Open Ephys Inc.). The electrode fixation device held flexible electrodes against the maxillary and mandibular gingiva while leaving the occlusal surface uncovered, allowing participants to chew physical samples during stimulation.

To prioritize real-time responsiveness, we adopted a threshold-based phase detection algorithm and used wired connections throughout the system to minimize communication delays. This design was motivated by the temporal constraints of multisensory integration, whereby sensory inputs must occur within a limited temporal window to be perceived as arising from a single event [29]. Although the temporal threshold depends on the sensory modalities involved, previous studies have suggested that the allowable delay for tactile feedback is on the order of 5–50 ms, with some studies arguing that delays should be kept below approximately 15 ms [30], [31]. Therefore, we used 15 ms as a conservative design target for phase-synchronized stimulation. The measured latency from sensor event detection to stimulation trigger output was less than 3 ms, which was well below this target. Figure 1 shows an overview of the chewing-synchronized electrical stimulation system.

B. Electrical Safety and Galvanic Isolation

Because the system delivers electrical stimulation inside the oral cavity while simultaneously acquiring sensor signals, electrical safety and circuit isolation were critical design requirements. We therefore implemented galvanic isolation between the low-voltage sensing/control circuit and the stimulation generation circuit, as shown in Fig. 2.

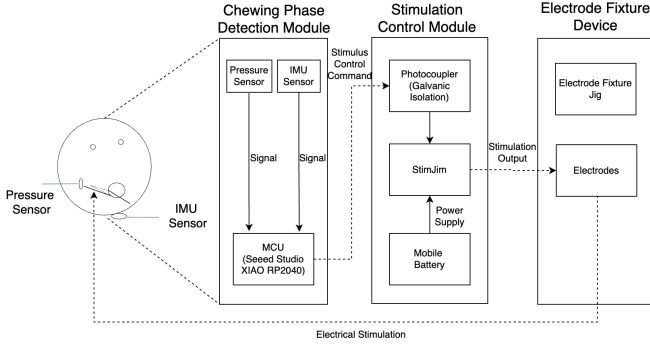


Fig. 1. Overview of the chewing-synchronized intraoral electro tactile stimulation system. Sensor signals from the pressure sensor or IMU are processed by the microcontroller, which sends trigger signals to the programmable stimulator. Electrical stimulation is then delivered to the gingival electrodes through the electrode fixation device.

The stimulation control module (StimJim) was powered by an independent mobile battery, thereby separating the stimulator ground from the measuring PC and the microcontroller. Trigger signals generated by the chewing phase detection module (XIAO RP2040) were transmitted to the StimJim input channels through photocouplers, maintaining electrical isolation between the control circuit and the stimulation circuit. In addition, two parallel control boards were used to independently control stimulation to the maxillary and mandibular gingiva, allowing stimulation intensity to be adjusted separately for each site because perceptual thresholds can differ across intraoral locations.

As a hardware-level safety measure, a normally closed (NC) emergency stop switch (A22NE-M-P102-N-B, Omron) was incorporated to allow stimulation to be stopped. Furthermore, to reduce the risk of short circuits caused by saliva along the wiring, a relay board was placed between the electrodes and the main device to provide additional physical separation and controlled switching of the stimulation pathway.

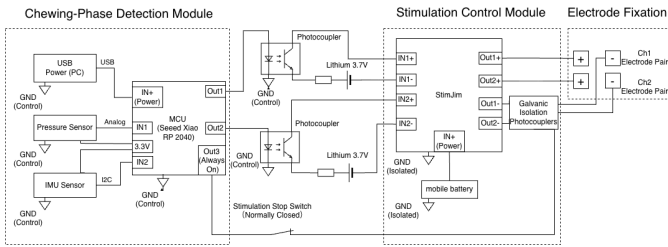


Fig. 2. Architecture of the chewing-synchronized intraoral electro tactile stimulation system. The sensing/control circuit and stimulation generation circuit are galvanically isolated using photocouplers. The StimJim is powered by an independent battery, and trigger signals are transmitted through isolated control paths. A normally closed emergency stop switch and relay board are integrated to interrupt stimulation and reduce the risk of short circuits caused by saliva along the wiring.

C. Real-Time Mastication Detection

The system was designed to trigger stimulation at two target mastication events, occlusion, defined as the timing of tooth-sample contact, and mouth opening, defined as the release

of contact. We used pressure-based detection for the silicone tube, whose repeated compression and recovery provided a consistent sensing cue, and IMU-based detection for the gummy candy, whose shape and mechanical state changed during mastication.

For trials involving a food-grade silicone tube (inner diameter: 1 mm; outer diameter: 3 mm), a pressure sensor (1 INCH-D1-4V-MINI, Amphenol) was connected to the tube. Biting the tube increased its internal pressure, allowing occlusion and opening to be identified when the signal crossed separate participant-specific upper and lower thresholds, respectively. The use of separate thresholds reduced unintended repeated triggers caused by small signal fluctuations near the detection boundary.

For trials involving a gummy candy (fruit gummy candy grape, Meiji), an inertial measurement unit (IMU; ICM-42688-P) was attached to the participant's chin using surgical tape (Transpore Surgical Tape, Solventum). Because the gummy candy could not be instrumented with the pressure sensor, the IMU was used to estimate mandibular movement during chewing. The gyroscope x-axis was aligned with the opening-closing direction of jaw motion and was therefore used for mastication-phase detection. Participant-specific thresholds were applied to the x-axis gyroscope signal to generate stimulation triggers corresponding to the target occlusion and opening events.

Before each experiment, detection thresholds were calibrated individually to account for differences in chewing rhythm, bite force, and jaw-motion amplitude. For calibration, participants performed several chewing cycles at an approximately regular rhythm of one cycle per second. The thresholds were adjusted until the trigger timing corresponded to the participant's perceived contact and release events. In addition, a software refractory period of 400–600 ms was introduced after each trigger to prevent multiple unintended triggers within a single chewing cycle.

D. Minimally Invasive Electrode Fixation Device

We designed an intraoral electrode fixation device to deliver electro tactile stimulation to the gingiva while leaving the occlusal surface uncovered. The device was required to satisfy three design requirements: leaving the occlusal surface uncovered, maintaining stable electrode contact with the gingiva, and reducing localized pressure on the oral mucosa. To provide texture-related oral feedback during chewing, the device was configured to stimulate the buccal and lingual gingiva of both maxilla and mandible on one side of the mouth.

The fixation frame consisted of silicone-coated piano and titanium wires connected to a 3D-printed elastic body that supported the electrodes. The piano wire (diameter: 0.7 mm; Daidohant Co., Ltd.) and titanium wire (diameter: 0.6 mm; Titanium Glint Co., Ltd.) served different mechanical roles. Based on pilot testing, the relatively stiff piano wire was used to provide buccal-lingual clamping force, whereas the more compliant titanium wire was formed into U-shaped clips that provided elastic recovery for maxillary fixation. Both wire types were coated with food-compatible silicone tubing and

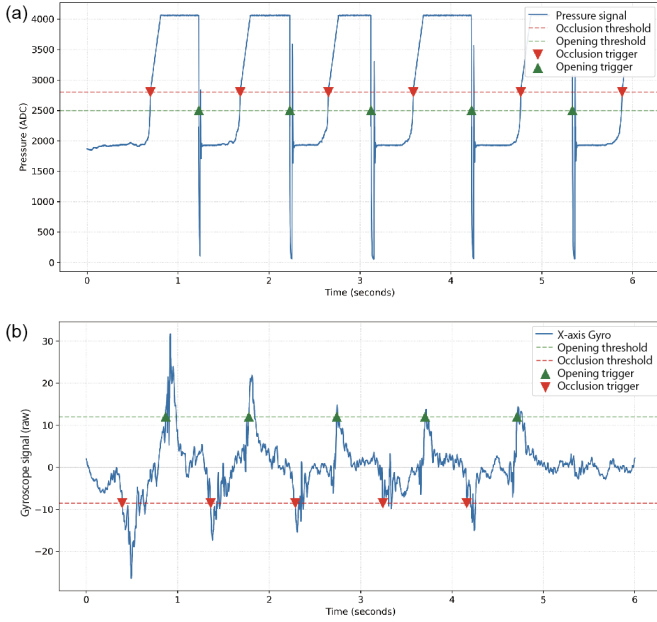


Fig. 3. Examples of real-time mastication-phase detection. (a) Pressure-based detection in the silicone tube condition. Occlusion and opening events were detected from threshold crossings in the pressure signal. (b) IMU-based detection in the gummy candy condition. Jaw opening and occlusion events were detected from the x-axis gyroscope signal. Red and green markers indicate occlusion and opening triggers, respectively. Dashed lines indicate the corresponding detection thresholds. The target events were biting onto the sample (occlusion) and releasing it during mouth opening. In each stimulation condition, a 100 ms stimulus was delivered following detection of the selected event. Occlusion-triggered and opening-triggered stimulation were tested in separate conditions.

food-grade silicone sealant (Food Equipment Silicone Sealant, AS ONE) to maintain fixation while reducing localized pressure on the oral mucosa.

To accommodate individual differences in oral size and shape, we fabricated multiple device sizes by varying the lengths of the wire arms. For the piano-wire arms, the total length after silicone coating ranged from 8.5 to 9.5 cm, corresponding to a core-wire length of 8 to 9 cm. The same length range was used for both the maxillary and mandibular devices. For the titanium-wire arms, the buccal-side length ranged from 8.5 to 9.5 cm, whereas the lingual-side length ranged from 8.0 to 8.5 cm. These arms were bent into U-shaped clips using a cylindrical object with a diameter of approximately 1 cm.

To reduce current density and localized discomfort, each electrode was designed with an enlarged contact area. The electrode shape consisted of a $16 \text{ mm} \times 8 \text{ mm}$ rectangle combined with a quarter circle of radius 8 mm, with corner fillets of radius 2 mm. The nominal geometric contact area of each electrode was approximately 178 mm^2 , before minor changes due to the corner fillets. A flexible flat cable (FFC) electrode assembly with four electrodes was fabricated to place electrodes on the buccal and lingual gingiva of both the maxillary and mandibular jaws. The spacing between paired electrodes was 10 mm, and the cable length was approximately 40 cm. Because perceptual thresholds for electrotactile stimulation can differ across intraoral locations, the maxillary and

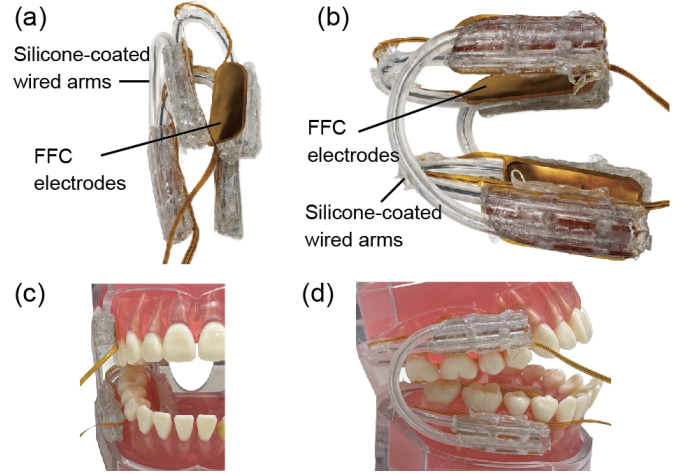


Fig. 4. Electrode fixation device. (a) Fabricated device viewed from the front, showing the FFC electrodes and silicone-coated wire arms. (b) Fabricated device viewed from the occlusal/open side, showing the paired upper and lower electrode modules. (c) Device mounted on a dental model in the frontal view, showing electrode placement on the buccal and lingual gingiva. (d) Device mounted on a dental model in the side view, showing that the occlusal surface remains uncovered during placement.

mandibular electrodes were controlled independently.

The 3D-printed body that supported the electrodes and connected the wire arms was fabricated using a biocompatible elastic resin (BioMed Elastic 50A, Formlabs). The body measured 5 mm in width, 10 mm in height, and approximately 28 mm in length. Its geometry included an electrode offset of approximately 3 mm. Two through-holes with a diameter of 3.2 mm were formed through the sides of the body to allow insertion of the silicone-coated wires. In addition, two adhesive injection holes with a diameter of 3.0 mm were formed from the top surface and connected internally to the side holes.

To improve conformity to the gingival surface, the curvature of the body was separately adjusted in Blender for the maxillary and mandibular versions. During assembly, silicone sealant was injected through the upper holes, and the silicone-coated wires were inserted through the side holes. Additional silicone sealant was then injected, and the assembled parts were left in a humid environment for more than 24 h to cure. The electrodes were attached to the body using a food-compatible instant adhesive (AS ONE) and silicone sealant, and the electrode leads were fixed to the wire frame using silicone adhesive.

Before the experiment, each participant selected the most suitable device size, and the wire arm angles were adjusted so that the device could be worn without discomfort.

IV. USER STUDY

We conducted a within-subjects user study to explore whether phase-dependent intraoral electrical stimulation can modulate texture-related perception during mastication.

A. Participants

Six participants completed the full experimental protocol and were included in the analysis. Fourteen participants were

initially recruited, and all underwent a device-fitting session before the experimental trials to confirm that the electrode fixation device could be worn safely and comfortably during chewing. Seven participants were excluded because the device did not fit their oral geometry, caused discomfort, or induced a gag reflex that prevented natural chewing throughout the experiment. One additional participant was excluded because they reported a strong stimulation-like sensation during the baseline no-stimulation condition, making it difficult to reliably attribute their subjective responses to the experimental stimulation. All six included participants were able to wear the device, chew the samples, and complete the subjective evaluations without reporting severe discomfort.

The final sample consisted of six male participants (mean age $\pm$ SD = 23.0 ± 1.1 years; range: 22–25 years). All participants self-reported having no history of oral sensory impairments, no dental treatment within the past three years on the same side of the mouth as the stimulation site, and no metal implants near the stimulation site. The experimental protocol was approved by the Institutional Review Board. All participants provided informed consent before participation.

B. Experimental Setup and Calibration

Before the main experiment, participants underwent a fit and safety screening procedure. After brushing their teeth and rinsing their mouths, an electrode fixation device was selected from several available sizes based on each participant's oral size and shape. If the device did not fit comfortably or stable electrode contact could not be achieved, the session was terminated at this stage.

Those who passed screening proceeded to calibration and the main experiment, with the electrode fixation device positioned around the maxillary and mandibular molar regions. Depending on participant comfort, the device was placed on either the right side ($n=5$) or the left side ($n=1$), with the electrodes positioned against the buccal and lingual gingiva.

After device fitting, two calibration phases were conducted. First, the stimulation current amplitude was calibrated separately for each frequency condition and electrode pair. The stimulation frequencies were 200 Hz and 1000 Hz, and the device included separate electrode pairs positioned on the maxillary and mandibular gingiva. Stimulation was delivered for 100 ms using a charge-balanced symmetric biphasic sinusoidal current waveform with no interphase gap. For each maxillary and mandibular electrode pair, the electrode assignment was fixed across participants: the lingual electrode served as the negative electrode and the buccal electrode served as the positive electrode during the initial phase of the waveform. Because the waveform was symmetric and biphasic, the instantaneous polarity alternated during the 100 ms stimulation period, and no DC component was applied.

For each electrode pair, the current amplitude was initially increased from 0 μ A in 100 μ A increments until the participant first reported perceiving the stimulation. The amplitude was then fine-tuned using smaller increments within the range of 1–10 μ A. Calibration was completed once the participant clearly perceived stimulation from both the maxillary and mandibular

electrode pairs without pain or discomfort. The amplitude was recalibrated during the experiment if the stimulation was no longer clearly perceptible.

Second, the mastication detection thresholds for the occlusion and opening phases were calibrated. Participants were asked to chew the sample at a steady pace of approximately 1 Hz while the sensor outputs were monitored and aligned with the intended trigger timings. The experiment consisted of two stages that differed in the sensor configuration and chewing sample.

In the first stage, a pressure sensor and silicone tube were used. For calibration of mastication-phase detection, the silicone tube was connected to the pressure sensor and formed into a U-shape, allowing participants to chew the tube at the region where electrical stimulation was perceived. The ends of the silicone tube remained outside the mouth during chewing. In this silicone tube condition, occlusion was detected when the pressure signal exceeded a predefined threshold, whereas opening was detected when the pressure signal fell below a predefined threshold. During calibration, participants were asked whether the stimulation was delivered at the perceived moment when the teeth contacted the sample in the occlusion condition or separated from it in the opening condition. If a temporal mismatch was reported, the corresponding threshold was adjusted manually, and the calibration was repeated until the stimulation was aligned with the intended mastication phase.

In the second stage, the chewing sample was changed to a gummy candy, and an IMU sensor was attached to the participant's chin. The IMU sensor was fixed using surgical tape (Transpore Surgical Tape, Solventum), with the gyroscope x-axis aligned to capture jaw rotation during opening and closing. Only the gyroscope x-axis signal was used for mastication-phase detection. During calibration, participants chewed the gummy candy at the region where electrical stimulation was perceived. Occlusion was detected when the angular velocity signal decreased below a predefined threshold, whereas opening was detected when the angular velocity signal exceeded a predefined threshold. As in the first stage, participants were asked whether the stimulation was delivered at the perceived moment when the upper and lower teeth came into contact or separated. If a temporal mismatch was reported, the corresponding threshold was manually adjusted, and the calibration was repeated until the stimulation was aligned with the intended mastication phase. The mastication-phase detection thresholds were recalibrated during the experiment whenever participants reported a mismatch between stimulation timing and their perceived chewing phase.

C. Procedure

The experiment consisted of two sessions that differed in the chewing sample and mastication-detection method. Participants first completed a food-grade silicone tube session using pressure-sensor-based detection, followed by a gummy candy session using IMU-based detection. This fixed order was used because the silicone tube provided more direct and stable detection of tooth-sample contact, allowing participants to

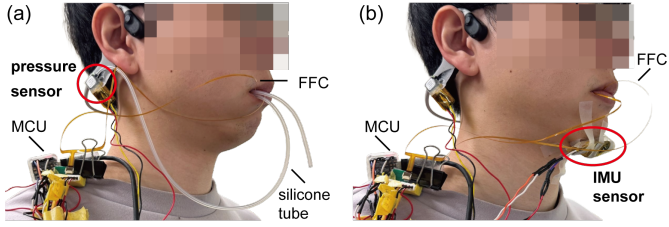


Fig. 5. Sensor configurations for mastication-phase detection. (a) Pressure-based detection in the silicone tube condition. (b) IMU-based detection in the gummy candy condition, in which participants chewed the gummy candy with the molars on the stimulated side. In both setups, sensor signals were processed by the MCU to trigger phase-dependent stimulation.

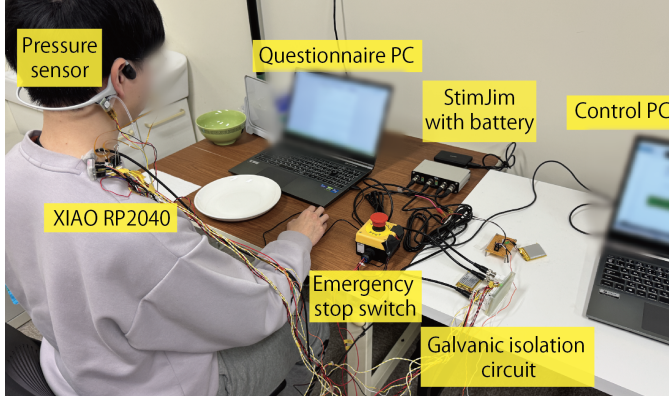


Fig. 6. Experimental environment. The participant wore the intraoral electrode fixation device and mastication-detection sensor while chewing the sample. Sensor signals were processed by the XIAO RP2040 and sent to the control PC. The XIAO RP2040 also sent trigger signals to the StimJim stimulator through a galvanic isolation circuit. A separate questionnaire PC was used for subjective ratings.

become familiar with the stimulation timing before proceeding to the gummy candy condition.

In each session, participants completed five stimulation conditions: baseline chewing without stimulation, 200 Hz stimulation during occlusion, 1000 Hz stimulation during occlusion, 200 Hz stimulation during mouth opening, and 1000 Hz stimulation during mouth opening. The order of these five conditions was randomized within each session. Each condition was repeated twice, resulting in 10 trials per session and 20 trials in total across the two sessions. During stimulation trials, electrical stimulation was delivered as a 100-ms sinusoidal current waveform whenever the target mastication phase was detected.

At the beginning of each trial, participants closed their eyes and positioned the sample on the stimulated side of the mouth. They were instructed to chew the sample five times while maintaining a steady rhythm as close as possible to 1 Hz. After each trial, participants completed the subjective evaluation items. A rest period of at least 30 s was provided between trials to reduce sensory adaptation and fatigue.

In the silicone tube session, participants chewed the U-shaped silicone tube, and its internal pressure was used to detect occlusion and opening events. In the gummy candy session, participants placed the gummy candy on the teeth near the stimulated gingival region and chewed it five times with

or without stimulation, depending on the assigned condition. After each gummy trial, participants spat out the sample without swallowing it. The experimenter then confirmed that no gummy residue remained in the mouth. When necessary, the device was removed, and both the mouth and device were cleaned before continuing the experiment.

D. Measurements

After each trial, participants completed a subjective questionnaire. Texture-related attributes, including hardness, elasticity, viscosity/adhesiveness, cohesiveness, and crunchiness, were rated on 7-point Likert scales. Stimulation-related perceptions, including incongruence, sense of integration, stimulation intensity, and perceived size of the chewed sample or evoked oral sensation, were also evaluated using 7-point scales. Examples and anchors were provided when necessary to clarify the meaning of each attribute, as summarized in Table I.

In addition to the rating items, participants provided qualitative descriptions of the evoked oral sensation. They reported sensation quality using multiple-choice and free-description responses, with examples such as vibration, pressure, or carbonation-like tingling. They also reported the perceived intraoral location, such as the gingival surface or inside the gingiva, and described the perceived spatial characteristics of the sensation, for example whether it felt round, sharp, or spread out. Participants could also provide free comments when they noticed other changes in texture-related impressions.

At the end of the experiment, we conducted a semi-structured interview to obtain detailed qualitative feedback. The interview addressed perceived changes in texture-related sensations, naturalness and integration with the chewed sample, device comfort and wearability, differences between the silicone tube and gummy candy sessions, perceived intraoral location of the evoked sensation, and possible improvements or applications of chewing-synchronized intraoral haptic feedback.

V. RESULTS

A. Calibrated Current Amplitudes

Stimulation was delivered using a symmetric biphasic sinusoidal current waveform, and intensity was quantified as the zero-to-peak current amplitude (μA). Before each session, the current amplitude was calibrated for each participant, frequency condition, and electrode pair. When participants reported that the stimulation was no longer clearly perceivable during the session, the amplitude was recalibrated to account for changes in electrode contact, saliva, or sensory adaptation.

To summarize the stimulation intensities used in the experiment, we calculated a representative amplitude for each participant, sample session, frequency condition, and electrode pair by averaging the initial and final calibrated amplitudes. We then calculated the mean and standard deviation of these representative amplitudes across participants.

Table II summarizes the calibrated current amplitudes. The 1000 Hz condition required higher amplitudes than the 200 Hz

TABLE I
QUESTIONNAIRE ITEMS RATED ON 7-POINT LIKERT SCALES. EXAMPLES AND ANCHORS WERE PROVIDED TO CLARIFY ITEM MEANINGS.

Item	Description	Examples / anchors
Hardness	Resistance felt when biting the sample with the teeth.	Tofu (1: soft), nuts (7: hard).
Elasticity	Rebound or springiness felt when the sample was pressed or chewed.	Gummy candy, mozzarella cheese.
Viscosity/adhesiveness	Sticky sensation or adherence to the teeth, gingiva, or oral cavity.	Mochi, caramel, peanut butter.
Cohesiveness	Degree to which the sample felt as if it remained together in the mouth during chewing.	Mochi, thick ham, steamed bun.
Crunchiness	Crispy or crunchy sensation perceived during chewing.	Potato chips, tempura, apple.
Incongruence	Degree to which the oral sensation felt unnatural or detached from the expected texture of the sample.	Sensation felt detached from the sample.
Stimulation intensity	Perceived intensity of the electrical stimulation.	1: very weak, 7: very strong.
Perceived size	Perceived size of the chewed sample or evoked oral sensation.	1: very small, 7: very large.
Sense of integration	Degree to which the evoked sensation felt integrated with the texture of the sample.	1: not integrated, 7: highly integrated.

TABLE II
REPRESENTATIVE CALIBRATED CURRENT AMPLITUDES. VALUES INDICATE MEAN $\pm$ SD ACROSS PARTICIPANTS.

Condition	Maxillary (μ A)	Mandibular (μ A)
Silicone, 200 Hz	735.8 $\pm$ 100.6	796.2 $\pm$ 199.5
Gummy, 200 Hz	752.5 $\pm$ 122.6	778.3 $\pm$ 201.6
Silicone, 1000 Hz	1016.2 $\pm$ 153.8	1041.2 $\pm$ 215.2
Gummy, 1000 Hz	1018.8 $\pm$ 125.8	1068.8 $\pm$ 157.6

condition across both sample sessions and electrode pairs. The mean representative amplitude ranged from 735.8 to 796.2 μ A for 200 Hz and from 1016.2 to 1068.8 μ A for 1000 Hz. Across both sample sessions and frequency conditions, the mandibular electrode pair tended to require slightly higher current amplitudes than the maxillary electrode pair. The overall representative amplitude ranges were 600–1130 μ A for 200 Hz and 800–1350 μ A for 1000 Hz.

B. Quantitative Subjective Evaluation

To evaluate changes in texture-related perception, we analyzed the 7-point Likert-scale responses. For each participant, the two repeated trials in each stimulation condition were averaged separately for each sample session. The silicone tube and gummy candy sessions were analyzed separately because they differed in both chewing sample and mastication-detection method.

For each questionnaire item, we conducted a non-parametric Friedman test to examine differences across the five stimulation conditions: Baseline, 200 Hz Occlusion, 1000 Hz Occlusion, 200 Hz Opening, and 1000 Hz Opening. Kendall's W was calculated as an effect-size measure for the Friedman test, with values of 0.1–0.3, 0.3–0.5, and greater than 0.5 interpreted as small, moderate, and strong effects, respectively. For items showing a significant main effect in the Friedman test, post-hoc pairwise comparisons were conducted using Wilcoxon signed-rank tests. As part of the exploratory analysis, uncorrected pairwise p -values were reported to characterize potential response patterns. These pairwise comparisons were interpreted as exploratory tendencies rather than confirmatory statistical evidence.

1) *Silicone Tube Sample*: In the silicone tube session, uncorrected Friedman tests yielded $p < .05$ for per-

TABLE III
STATISTICAL SUMMARY OF QUESTIONNAIRE ITEMS WITH UNCORRECTED FRIEDMAN $p < .05$. ONLY THESE ITEMS ARE SHOWN.

Session	Item	Friedman p	Kendall's W
Silicone	Stimulation intensity	.0075	.580
Silicone	Incongruence	.0056	.609
Silicone	Viscosity/adhesiveness	.0039	.642
Silicone	Crunchiness	.0047	.626
Silicone	Perceived size	.0291	.449
Gummy	Stimulation intensity	.0049	.622
Gummy	Incongruence	.0033	.659
Gummy	Elasticity	.0267	.458

Note. Pairwise post-hoc comparisons were uncorrected and are shown as exploratory annotations in Figs. 7 and 8.

ceived stimulation intensity, perceived incongruence, viscosity/adhesiveness, crunchiness, and perceived size (Table III; Fig. 7). The texture-related responses showed exploratory condition-dependent patterns, with higher viscosity/adhesiveness ratings in both opening-phase conditions than in baseline and higher crunchiness ratings in the 1000 Hz occlusion condition than in baseline and both opening-phase conditions, based on uncorrected pairwise comparisons. Perceived size also showed an exploratory increase during stimulation. Incongruence and stimulation intensity tended to be higher in stimulation conditions than in the no-stimulation baseline, consistent with the electrical stimulation being clearly perceived.

2) *Gummy Candy Sample*: In the gummy candy session, uncorrected Friedman tests yielded $p < .05$ for stimulation intensity, incongruence, and elasticity (Table III; Fig. 8). The elasticity result suggested a condition-dependent pattern, whereas incongruence and stimulation intensity mainly reflected the presence of electrical stimulation. Compared with the silicone tube session, fewer texture-related items showed condition-dependent variation. Because the two sessions differed in both chewing sample and mastication-detection method, this difference should be interpreted cautiously.

C. Qualitative Feedback

The qualitative feedback helped interpret the questionnaire patterns while also revealing substantial individual variability. Participants did not uniformly interpret the stimulation as a change in food texture. Instead, their comments suggested two competing interpretations. In some trials, the electrically evoked sensation was related to texture-like changes in the

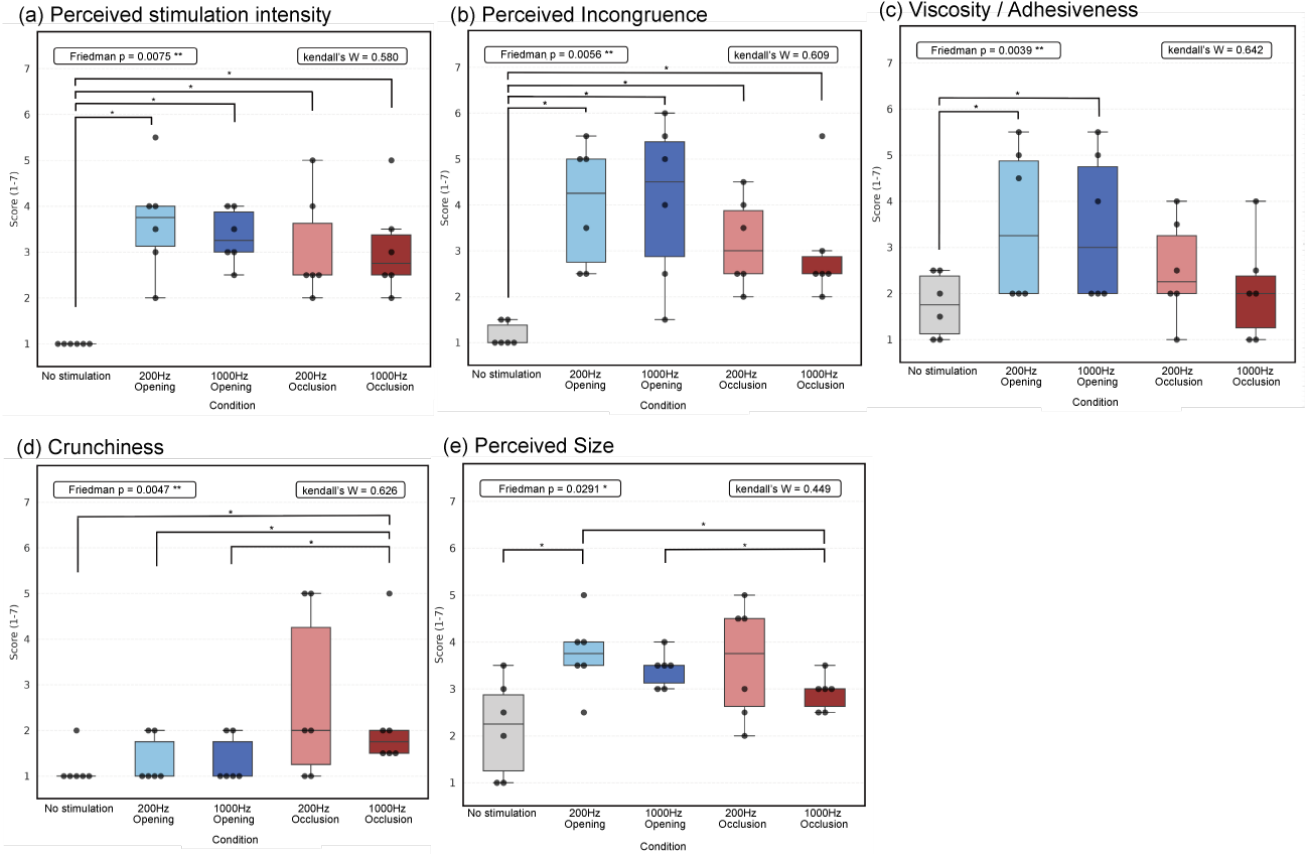


Fig. 7. Questionnaire results for the silicone tube session. Boxplots show ratings for (a) perceived stimulation intensity, (b) perceived incongruence, (c) viscosity/adhesiveness, (d) crunchiness, and (e) perceived size across the five stimulation conditions. Dots indicate individual participants. Brackets indicate uncorrected exploratory post-hoc comparisons (* $p < .05$, ** $p < .01$); they should not be interpreted as confirmatory multiple-comparison-corrected effects.

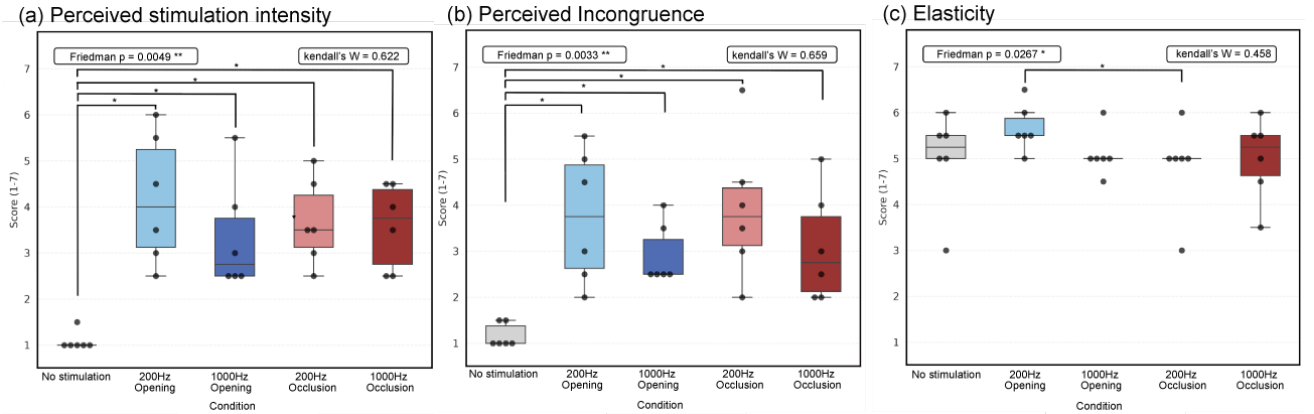


Fig. 8. Questionnaire results for the gummy candy session. Boxplots show ratings for (a) perceived stimulation intensity, (b) perceived incongruence, and (c) elasticity across the five stimulation conditions. Dots indicate individual participants. Brackets indicate uncorrected exploratory post-hoc comparisons (* $p < .05$, ** $p < .01$); they should not be interpreted as confirmatory multiple-comparison-corrected effects.

chewed sample, whereas in other trials it was perceived as an artificial electrical sensation separate from the sample.

Several comments were consistent with texture-like interpretations. In the silicone tube session, participants sometimes described the sample as feeling “thicker”, “wider”, or “larger and round” during stimulation, which was consistent with the exploratory increase in perceived size. Adhesive or viscous impressions were also reported. For example, some

participants described opening-phase stimulation as producing a sticky or mochi-like feeling, as if the sample remained attached to the teeth after biting. Comments related to hardness, roughness, or crunchiness were also observed, particularly when stimulation occurred during tooth-sample contact. These reports were broadly consistent with the questionnaire pattern in which opening-phase stimulation was associated with viscosity/adhesiveness and occlusion-phase stimulation

was associated with crunchiness in the silicone tube session.

At the same time, many comments indicated that the stimulation was not experienced as a natural texture change. Participants often described the sensation using terms such as tingling, buzzing, pressure on the gingiva, carbonation-like sensation, or electrical stimulation. Some comments also indicated that the stimulation felt incongruent with chewing, was perceived as delayed relative to the chewing event, or masked the original texture of the sample. These reports suggest that the texture-related ratings may have been influenced not only by texture-like interpretation, but also by the salience and artificiality of the electrical stimulation itself.

The gummy candy condition produced more mixed responses than the silicone tube condition. Some participants described the stimulation as being added on top of the gummy texture, whereas others reported little change or difficulty distinguishing the stimulation from the original chewing sensation. Several comments also suggested that device fit, tongue movement, and the physical texture or taste of the gummy candy affected how clearly the stimulation was perceived. Overall, the qualitative feedback supports a cautious interpretation: chewing-synchronized gingival electrotactile feedback can sometimes be interpreted as texture-related, but this interpretation depends on stimulation timing, stimulus salience, device comfort, and the chewing sample.

1) *Trial-by-Trial Free Descriptions*: The trial-by-trial free descriptions provided additional qualitative evidence for condition-dependent changes in texture-related impressions. In the no-stimulation condition with the silicone tube, most participants described the sample using expressions consistent with its actual shape, such as “thin cylinder” or “flat”. In contrast, during 200 Hz electrical stimulation, participants used descriptions such as “thick”, “larger and round”, and “wide”, suggesting changes in the apparent shape or spatial extent of the chewed sample. This observation is consistent with the pattern observed in perceived-size ratings.

Several comments also suggested a relationship between stimulation timing and adhesive impressions. P5 commented that stimulation occurring after the bite produced a somewhat sticky sensation, and P6 described the sensation as “sticky”. In the final interview, P3 reported that stimulation delivered after biting made the sample feel adhesive and suggested that a longer delay might further enhance this impression. Together, these comments suggest that stimulation occurring after the bite could be interpreted in relation to adhesion or residual contact with the sample.

In addition, some participants associated high-frequency stimulation with hardness. P3 and P6 commented that the 1000 Hz stimulation felt somewhat hard. Comments related to crunchiness were also observed. In the silicone tube condition, P6 gave the highest crunchiness rating for the 200 Hz occlusion condition and described the perceived shape as “multi-layered”. Similarly, P3 reported that the 200 Hz occlusion condition produced a rough or crunchy sensation. These comments are broadly consistent with the quantitative patterns observed for texture-related impressions such as adhesiveness and crunchiness, while also suggesting a possible association between stimulation frequency and perceived hardness.

a) *Naturalness and Wearability Limitations*: The naturalness of the modulated texture varied among participants. P2 mentioned, “When the stimulation matched the exact timing of the bite (occlusion), it felt natural. Weaker stimulation resulted in less unnaturalness”. However, participants (P1, P3, P6) who strongly perceived the texture-related changes also noted the physical limitations of the device. The physical presence of the device and prolonged wearing time (approximately 2 hours) were reported to cause localized pressure on the gingiva, occasionally mask subtle electrical cues, and hinder natural jaw movements, particularly with the gummy sample, for which natural tongue movement is important.

VI. DISCUSSION

A. The Role of Dynamic Context in Multisensory Integration

The present results do not demonstrate robust food texture rendering. Instead, they suggest that, in the silicone-tube condition, the timing of gingival electrotactile feedback was associated with shifts in selected texture-related ratings, especially viscosity/adhesiveness and crunchiness. This differs from previous studies, in which electrically evoked intraoral sensations were described primarily in terms such as vibration, tingling, or pressure [17], [18].

This contrast suggests that the perceptual interpretation of an electrically evoked oral sensation may depend on the dynamic context in which it is delivered. Texture perception is fundamentally a multisensory integration process. When electrical stimulation is applied within the dynamic context of actual mastication, participants may integrate the artificial tactile input with natural kinesthetic and proprioceptive feedback from the jaw muscles and temporomandibular joint. This contextual anchoring may have led participants to interpret the electrical stimulation less as isolated oral sensations and more as sensations related to the chewed sample.

B. Temporal Context in Interpreting Intraoral Tactile Feedback

A key observation from this study is that the perceived meaning of intraoral electrotactile feedback appeared to depend on the chewing phase in which it was delivered. Natural tooth loading is partly encoded by periodontal mechanoreceptors, including Ruffini-like endings, which contribute to sensing force amplitude and direction during mastication [6]–[9]. However, the present stimulation was delivered through gingival electrodes near the molar region and should not be interpreted as selective activation of the periodontal ligament. Unlike natural tooth loading, the electrical stimulus did not reproduce a specific mechanical deformation pattern of the tooth, gingiva, or periodontal tissues. The stimulation likely activated multiple sensory afferents within the electric field, potentially including those arising from the periodontal ligament, thereby producing an artificial oral tactile sensation whose perceptual interpretation depended on the ongoing chewing context.

This distinction is important for interpreting the present results. The observed changes in viscosity/adhesiveness, crunchiness, and perceived size should not be taken as evidence

that the system reproduced natural periodontal force feedback. Rather, the results suggest that an electrically evoked oral sensation could be interpreted differently depending on whether it occurred during tooth-sample contact or during mouth opening. During occlusion, the sensation may have been associated with contact, resistance, or fracture-like events. During opening, it may have been associated with separation, adhesion, or residual contact with the sample.

These exploratory findings suggest that temporal context should be considered when designing intraoral haptic feedback intended to influence texture-related impressions during chewing. Rather than acting as a standalone tactile stimulus, gingival electrotactile feedback may acquire texture-related meaning when it is embedded in the ongoing sensorimotor context of mastication. At the same time, because many participants also described the stimulation as electrical, artificial, or incongruent, the present results should be interpreted as timing-related biases in texture-related impressions rather than as evidence of robust or naturalistic food texture rendering.

C. Limitations and Future Work

This study has several limitations. First, the sample size was small, and the statistical analyses involved multiple uncorrected tests. The quantitative results should therefore be interpreted as exploratory patterns rather than confirmatory evidence. A larger study with corrected multiple comparisons is needed to verify the robustness of the observed effects.

Second, the silicone tube and gummy candy sessions differed not only in the chewing sample but also in the mastication-detection method. The silicone tube session used pressure-based detection, whereas the gummy candy session used IMU-based detection. The session order was also fixed, with the silicone tube session always preceding the gummy candy session. Consequently, differences between the two sessions cannot be attributed solely to the sample material and may also reflect differences in sensing method, familiarization, adaptation, or fatigue. Future studies should separate these factors using a counterbalanced and more controlled experimental design.

Third, the final sample was limited to six male participants in their twenties. The generalizability of the findings should therefore be examined with a larger and more diverse participant group. In addition, individual chewing habits, dominant chewing side, bite force, and jaw-motion patterns were not strictly controlled. Future systems should incorporate more precise sensing of chewing kinematics and bite force to adapt stimulation timing to individual chewing behavior.

Finally, device fit was a major practical limitation. Eight of the fourteen recruited participants were excluded before analysis because of device fit, discomfort, gag reflex, or unreliable baseline responses. This high exclusion rate indicates that the current fixture design is not yet suitable for broad deployment and should be regarded as a laboratory prototype for exploratory evaluation. Even among participants who completed the study, the physical presence of the device could still mask subtle electrical cues or interfere with natural jaw and tongue movements. Future device iterations should improve

miniaturization, fit, and long-term wearability, for example by using thinner flexible electrode structures or personalized designs.

VII. CONCLUSION

This paper presented an exploratory study of chewing-synchronized gingival electrotactile stimulation for modulating selected texture-related subjective ratings during active chewing. We developed a low-latency stimulation system that detects mastication events and delivers gingival electrical stimulation during either occlusion or mouth opening while preserving tooth-sample contact. We also designed an intraoral electrode fixation device that allowed stimulation of the maxillary and mandibular gingiva without covering the occlusal surface.

The user study suggested that the interpretation of intraoral electrotactile feedback varied across stimulation conditions and chewing phases. In the silicone tube condition, opening-phase stimulation was associated with increased viscosity/adhesiveness, whereas occlusion-phase stimulation was associated with increased crunchiness. Perceived size and stimulation-related ratings such as incongruence and intensity also differed across stimulation conditions. In the gummy candy condition, fewer texture-related differences were observed, although elasticity showed an exploratory condition-dependent pattern. Qualitative feedback further suggested that temporally aligned stimulation could be interpreted in relation to the chewed sample rather than only as an isolated electrical sensation.

These exploratory findings suggest that gingival electrotactile feedback can be used as a temporally embedded oral tactile cue for studying how chewing context shapes the interpretation of artificial intraoral sensations. However, the present results should be interpreted as initial evidence of timing-related subjective biases rather than as a demonstration of robust food texture rendering. Future work should improve device fit, increase participant diversity, introduce counterbalanced and asynchronous control conditions, and separate the effects of chewing sample, detection method, and stimulation timing.

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